Ezitech Engineering Framework (EEF)

Industry ML Case Study – ML-003

Enterprise Healthcare Early Disease Risk Prediction & Clinical Decision Support Platform

Duration

4 Weeks

Team Size

2 ML Engineers

Industry Background

Hospitals generate millions of patient records every year, including laboratory results, vital signs, medical history, prescriptions, imaging reports, and clinical observations.

Doctors often need to identify high-risk patients before symptoms become critical. Traditional diagnosis depends heavily on manual assessment and may delay early intervention.

Healthcare providers require an AI-powered clinical decision support system capable of predicting disease risks and assisting medical professionals with evidence-based recommendations.

Business Problem

A healthcare network operates:

- 80 Hospitals
- 2,500 Doctors
- 20 Million Patient Records
- 5 Million Annual Lab Reports

Current challenges:

- Late disease detection
- ICU overcrowding
- High readmission rates
- Manual patient prioritization
- Limited predictive insights
- Increasing treatment costs

The organization requires an intelligent platform capable of predicting patient risk before conditions become severe.

Project Objective

Develop an **Enterprise Healthcare Early Disease Risk Prediction & Clinical Decision Support Platform** that analyzes patient records, laboratory results, vital signs, and historical diagnoses to estimate disease risk, prioritize patients, and provide explainable recommendations for clinicians.

Core Modules

Patient Data Pipeline

Collect:

- Demographics
- Medical History
- Vital Signs
- Laboratory Results
- Medication Records
- Previous Diagnoses
- Lifestyle Factors
- Family History

Feature Engineering

Generate:

- BMI

- Blood Pressure Trends
- Blood Sugar Trends
- Heart Rate Variability
- Cholesterol Ratios
- Kidney Function Scores
- Risk Indexes
- Medication Compliance Features

Disease Prediction Engine

Predict risks for:

- Diabetes
- Heart Disease
- Stroke
- Kidney Disease
- Liver Disease
- Hypertension

(Datasets can initially focus on one disease and be designed for future expansion.)

Patient Risk Scoring

Generate:

- Low Risk
- Moderate Risk
- High Risk
- Critical Risk

Each patient receives:

- Probability Score
- Confidence Score
- Severity Level

Explainable AI

Display:

- Top Contributing Features
- SHAP Values
- Feature Importance
- Prediction Confidence
- Clinical Explanation

Clinical Dashboard

Provide:

- High-Risk Patient List
- Disease Distribution
- Population Health Trends
- Department Statistics
- Risk Heatmaps

Alert System

Notify clinicians when:

- Critical Risk Detected
- Sudden Vital Changes
- Abnormal Lab Results
- High Readmission Risk

Machine Learning Components

Implement:

- XGBoost

- XGBoost
- LightGBM
- Random Forest
- Logistic Regression (Baseline)
- Ensemble Learning
- Feature Selection
- Hyperparameter Optimization

MLOps Components

Include:

- MLflow
- Model Registry
- Dataset Versioning
- Automated Retraining
- Data Drift Detection
- Performance Monitoring

Technical Requirements

Backend

- Python
- FastAPI
- PostgreSQL

ML Stack

- Scikit-learn
- XGBoost
- LightGBM
- MLflow
- Pandas
- NumPy
- SHAP

Non-Functional Requirements

Support:

- Secure API Access
- Explainable Predictions
- Horizontal Scalability

- High Availability
- Regulatory-Friendly Logging
- Low-Latency Inference

Deliverables

- Complete Source Code
- ML Pipeline
- Feature Engineering Pipeline
- Prediction API
- Clinical Dashboard
- Model Evaluation Report
- Architecture Diagram
- Deployment Guide
- README
- Technical Presentation

Bonus Challenges

Implement one or more:

- Medical Image Integration (future-ready)

- LLM-Based Clinical Report Summarization
- Personalized Treatment Recommendation Engine
- Time-Series Patient Monitoring
- Wearable Device Integration

Evaluation Criteria

Criteria	Weight
Feature Engineering	20%
Model Performance	20%
Explainability	20%
MLOps	15%
Clinical Dashboard	10%
Documentation	10%
Presentation	5%

Expected Learning Outcomes

By completing this case study, interns should demonstrate the ability to:

- Build healthcare-focused predictive ML systems.
- Engineer clinical features from structured patient data.
- Develop explainable AI suitable for medical decision support.
- Implement production-ready MLOps pipelines.
- Deliver enterprise-grade healthcare analytics solutions.

Business Impact

This platform helps hospitals identify high-risk patients earlier, enabling timely intervention and better resource allocation. By combining predictive analytics with explainable AI, clinicians gain actionable insights while maintaining transparency in decision-making. The system supports improved patient outcomes, reduced treatment costs, and more efficient healthcare operations.